

Market power, optimal investment and sustainable economic growth in Kazakhstan

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Abstract: This study explores optimal capital stocks and investment for Kazakhstan in a neoclassical growth model that has been extended to account for the role of market power, fixed costs, and excess profits. Our study is distinct in examining these issues in an emerging economy and at a disaggregated level. We apply a neoclassical growth model extended to incorporate imperfect competition, where markups could reflect either fixed costs of production or market power generating excess profits. We focus on estimating the model for 2016-2023 and 2023. We calibrate the model at the aggregate level and for 10 1-digit NACE industries, utilizing confidential data from Kazakhstan's Bureau of National Statistics to estimate sectoral markups. We use results from 2-digit NACE industries and 1-digit industries from Australia, Norway, and Nigeria to help interpret results. Our analysis reveals that market power is important for recognizing the higher marginal product of capital in mining and information technology (IT), which both have substantial overheads. We find no evidence of overinvestment for the aggregate economy, but our 2016-2023 results suggest overinvestment in agriculture and utilities, while our 2023 results suggest recent overinvestment in manufacturing. The information technology sector has substantial underinvestment. We estimate that Kazakhstan could achieve an optimal capital stock in five years with investment at current levels, but better investment allocation across industries would reduce necessary aggregate investment by 10%. In addition to horizontal policies to improve financial market development and investment attractiveness, our results support policies in the IT sector to facilitate investment and increase competition among medium- and large-sized firms, as well as policies to improve the agriculture and manufacturing sectors' ability to absorb and allocate investment.

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1. Introduction

This paper presents our research findings on the optimal level of investment required to achieve long-term sustainable economic growth in Kazakhstan, while taking a differentiated approach that considers overinvestment and underinvestment by industry. This research offers a data-driven approach that can assess Kazakhstan's present investment allocation and set priorities for improving industrial policy.

Our approach is to assess the marginal product of capital (MPK) by industry using a standard neoclassical growth model that has been augmented to account for the possibility of imperfect competition and market power (Ball and Mankiw, 2023). Concerns about imperfect competition and increasing markups have escalated recently (Eeckhout & De Loecker, 2020), including in Kazakhstan (De Groot & Konings, 2023). We focus on industry parameters calibrated using data for 2016-2023 and 2023. The model can be calibrated with industry parameters that take into account variation in distinct features including capital intensity, depreciation, and the extent of imperfect competition. The model yields distinct MPK by sector, which is common in emerging economies with underdeveloped capital markets (David & Venkateswaran, 2019; Bau & Matray, 2023). The baseline model can immediately offer insights into overinvestment and underinvestment, according to a Phelps (1961) Golden Rule criteria such that investment optimizes consumption. With standard assumptions about decreasing returns, the model implies optimal capital stocks by sector.

For the aggregate economy, our model suggests Kazakhstan's capital stock could be about 90% higher, so there is substantial room for further growth through investment without additional improvement in total factor productivity (TFP). The

optimal capital stock could be achieved in roughly five years, if investment grows similarly in proportion to existing capital stocks, but recent patterns of investment are quite distinct, with substantial additional capital devoted to manufacturing for the purposes of industrial diversification.

Our sectoral analysis reveals misallocation in investment across industries. Our model suggests consistent overinvestment in utilities and agriculture and more recent overinvestment in manufacturing. Information technology (IT) exhibits substantial underinvestment – too little investment to cover depreciation and keep the capital stock stable, despite high returns, suggesting further room to improve policy supporting the IT sector. Our approach using market power is particularly important for assessing the IT sector. Improving investment allocation across sectors would achieve an efficient economy-wide MPK with 10 percent less investment necessary than allocating investment in proportion to current capital stocks. Although the importance of economic diversification to Kazakhstan's macroeconomic stability is noted, our results suggests that investment in agriculture and manufacturing has yielded subpar returns to date, suggesting adjusting policy to allow better absorption or allocation of capital.

The report proceeds as follows, according to the required structure. We next review key literature to justify our approach and offer background on policy approaches to investment (Section 2). Third, we discuss our methodology more formally, including an initial discussion of its strengths and limitations (Section 3). Fourth, we detail data used for the analysis (Section 4). Fifth, we discuss our analysis of the marginal product of capital for the aggregate economy and the economy divided into sectors (Section 5). Sixth, we discuss our results on optimal

capital stock and the optimal level of investment by sector over a 5-year plan that yields such a stalk (Section 6). Lastly, we present recommendations for policy.

2. Review of Scientific Research on Optimal Investment and Related Policy

This section first discusses the theory of marginal product of capital most relevant to our analysis, and then we summarize lessons for Kazakhstan’s policy from other related literature. Further specifics on select studies are provided in Appendix A.

2.1 Economic Theory on Marginal Product of Capital and Empirics

A starting point of our analysis is that investment is amenable to analysis at the levels of the macroeconomy, the industry, and the firm, such as the Konings et al. (2022) analysis of underinvestment. The focus of our report is on overinvestment vs. underinvestment at the macroeconomic level and the industry level, while we see our industry findings as suitable for motivating more targeted industry case studies with firm-level analysis in future research. For research with policy recommendations for development based on firm-level data according to firm size and ownership, we refer to World Bank research such as Iacovone et al. (2025).

Our approach builds on the core idea of the Golden Rule of Phelps (1961), which relates to the MPK. The Golden Rule, based on a Solow (1956) growth model with constant returns to scale of production and an exogenous saving rate, proposes that too much savings can lead to an economy with too much investment that leads to a suboptimal consumption. The Golden Rule trades off consumption against capital per worker with decreasing returns, and at the optimum, $f' - \delta = (n + g)$, where f' is the MPK in terms of capital per worker, δ is depreciation, n is the population growth rate, and g is the exogenous growth rate per capita. The term $f' - \delta$ is the *social or net return to capital*. An economy with more capital than this

is *dynamically inefficient* (Abel et al., 1989) since less investment could increase consumption. An economy that is not dynamically inefficient is *dynamically efficient* though not necessarily behaving optimally. The recent work by Ball and Mankiw (2023) extends this concept to settings with imperfect competition, where the Golden Rule still holds, but methods of deriving the MPK from data are distinct.

To derive the MPK from data, a common approach is $MPK = \alpha Y/K$, where α is the capital share of income, which holds under an assumption of constant returns to scale. This is the starting point for empirical measurement of the MPK (Caselli & Feyrer, 2007) in cross-country comparisons, such as the Lucas (1990) paradox which asked why more capital does not flow to poor countries with high MPKs. The Ball and Mankiw (2023) extension, in brief, argues that the MPK needs to be adjusted upward when there are fixed costs. The MPK is higher when there than $\alpha Y/K$ when there is a markup from imperfect competition, but this needs to be adjusted back down based on the amount of the markup that flows to pure or excess profits rather than production, and the result is a MPK that accounts for fixed costs.

We remark here that the idea of market power and fixed costs is extremely general and can apply to any industry, including in Kazakhstan. Perfect competition requires that prices are set to marginal costs. Imperfect competition allows prices above marginal costs. Some markups may flow to excess profits rather than fixed costs. This can be observed in accounting data for any industry anywhere, in principle.

Looking beyond the macroeconomic level, there is potential for misallocation of capital across firms and industries. This literature focuses on misallocation across firms and often emphasizes financial market underdevelopment as a reason (David & Venkateswaran, 2019; Bau & Matray, 2023) for misallocation. There is surprisingly little focus on using MPK to assess misallocation at the industry level.

2.2 Policy Insights for Kazakhstan from prior academic economic literature

Kazakhstan must make strategic investments in several important areas, such as infrastructure, agriculture, and research and development (R&D), while tackling underlying institutional and governance issues if it hopes to achieve long-term, sustainable economic growth. Yazgan and Yalcinkaya (2018) emphasize how crucial it is to boost R&D spending, especially in the private sector and academic institutions, in order to promote technological innovation and the development of human capital, both of which are essential for Kazakhstan's economic diversification. To get the most out of these investments, though, this needs to be combined with improved sector coordination and institutional support. In a similar vein, Gollin et al. (2002) and Badibanga and Ulimwengu (2020) stress the necessity of modernizing agriculture through greater investment, which can act as a basis for more extensive economic change even though agriculture is no longer the major sector. To guarantee that these investments result in significant growth, governance concerns must be resolved and infrastructure must be improved.

Additionally, the literature emphasizes how structural change can speed up economic growth. McMillan et al. (2014) point out that transitions from low- to high-productivity sectors can be sluggish, particularly when there is inadequate governance. The slow pace of Kazakhstan's shift from resource-dependent to more diversified industries highlights the necessity of governance reforms to promote

sustainable growth. Furthermore, models by Abel (1983) and Brock and Mirman (1972) highlight how important it is for Kazakhstan to manage volatility in international commodity markets, especially given its reliance on the price of gas and oil. Preventing overinvestment or underinvestment in vital industries requires maintaining balanced investment levels and reducing the impact of market volatility. Lastly, De Loecker et al. (2020) emphasize that in order to guarantee that investments result in dynamic, inclusive growth, regulations that encourage competition and lessen market concentration are necessary, especially in the energy sector. All things considered, Kazakhstan has to combine market-oriented policies, institutional changes, and targeted investments to achieve sustainable growth.

2.3 Insights for Kazakhstan from Policy Institutions

According to Gelb (2010) and Deléchat et al. (2024), horizontal policies—like enhancing trade openness, human capital development, and macroeconomic stability—are essential for reducing Kazakhstan's resource-dependent economy's instability. According to Deléchat et al. (2024), these policies are immediately pertinent since they support Kazakhstan's need to diversify its economy and bolster non-resource sectors, especially through infrastructure investment. To prevent market inefficiencies and encourage diversification, Kazakhstan must also concentrate on vertical policies that target certain industries, such as manufacturing and renewable energy.

For Kazakhstan to overcome challenges such as corruption and resource curse which are compatible with Kazakhstan's reality Pouokam (2023) emphasizes the significance of good governance and innovation, addressing problems like political instability and corruption. Her suggestions for institutional changes align

with Kazakhstan's continuous initiatives to improve openness and lessen its dependency on oil earnings. Similar to this, Gelb, Tordo, and Halland (2014) and Berg et al. (2013) support the use of Sovereign Wealth Funds (SWFs) to finance long-term growth and stabilize the economy. The National Fund of the Republic of Kazakhstan is a tool used to balance public investment and support infrastructure development. Moreover, the focus on counter-cyclical fiscal policies by Izvorski et al. (2018) is especially relevant for Kazakhstan, whose economy is subject to large swings as a result of the volatility of the world oil price. By putting these strategies into practice, Kazakhstan may be able to enhance fiscal stability and guarantee sustainable investments throughout recessions. All things considered, the literature provides pertinent tactics; Kazakhstan's difficulty is in tailoring these methods to its particular political, social, and economic circumstances.

2.4 Literature on Underinvestment and Overinvestment

According to La Rocca et al. (2005), disagreements among managers, owners, and debtholders frequently lead to less-than-ideal investment choices. The high ownership concentration in family-owned and state-controlled companies in Kazakhstan may lessen the incentives for excessive investment, but it may also result in inefficiencies. The danger of overinvesting in non-strategic projects could be reduced by coordinating managerial incentives with long-term value generation, for example, through performance-based compensation. He and Kondor (2016) also point out that financially strapped businesses typically underinvest in recessions and overinvest in booms, a cyclical trend that is particularly pertinent to Kazakhstan's resource-driven economy. Effective policy management of this boom-bust cycle can assist stabilize investments by limiting

overspending during periods of high oil prices and guaranteeing steady investment during periods of decline.

Furthermore, Ho et al. (2017) used the Red Queen competition paradigm, which implies that deviating from industry investment standards might either improve or worsen firm performance. This emphasizes the necessity for Kazakhstan to balance investments across industries such as technology, agriculture, and energy in order to prevent stagnation from underinvestment or declining returns from overinvestment. Finally, Mishra (2024) emphasizes the value of financial tools like loan guarantees and subsidized loans to reduce underinvestment, especially in agriculture, a sector that is essential to Kazakhstan's diversification. Targeted financial strategies are necessary to revitalize agriculture and promote overall economic development, even though it is no longer a main industry due to its reduction in export volume.

3. Research methodology and limitations

This section discusses the theoretical framework of Ball and Mankiw (2023), which we build upon and extend in the context of Kazakhstan. First, we present the structure of the model, including its key assumptions and main economic mechanisms. Second, we explain how the model is implemented using macroeconomic data for Kazakhstan, focusing on the calibration of parameters and measurement of key relationships. Third, we discuss our extension to a multi-industry setting. Lastly, we discuss some limitations of the framework in our context, which will be essential in interpreting our results and developing reasonable policy recommendations.

3.1 The Ball and Mankiw (2023) Model

We present the main components of the Ball and Mankiw (2023) framework, which extends the standard neoclassical growth model to account for imperfect competition and market power. This model introduces a wedge between the marginal product of capital and the real interest rate and allows for the evaluation of dynamic efficiency in this setting.

3.1.1 Core assumptions

The model assumes an economy composed of N identical firms, each producing and selling a differentiated good. Each firm has market power, meaning it sets a markup on its prices over marginal cost. Firms operate under imperfect competition, which allows them to determine output and pricing independently. Regarding the production function, producing one unit of the firm's output requires one unit of a composite of capital (K) and labour (L) and λ units of intermediate goods. Production is subject to labor-augmenting technological progress, with the level of technology A growing at an exogenous rate g . At the same time, labour is inelastically supplied and grows at rate n . The economy saves an exogenous fraction of output, and capital depreciates at rate δ .

3.1.2 Microfoundations and Price Setting

Each firm's production technology is:

$$Q = F[K, A(L - X)],$$

where K is capital, L is labour, A is labour-augmenting technology, X is fixed overhead labour. The production function $F(\cdot)$ is homogenous of degree one, indicating constant returns to scale with respect to inputs. If $X=0$, firms exhibit constant returns to scale (similar to the Solow growth model). If $X>0$, then firms

exhibit increasing returns to scale. Intermediate goods are not necessary for the core theory but are included to ensure the calibration aligns with real-world data.

Regarding markups, each firm has market power and charges a markup γ over marginal cost:

$$P = \gamma MC,$$

and the model assumes that $\gamma\lambda < 1$. Each firm hires labour at rate W and rents capital at rate R in competitive factor markets. It also buys goods from other firms, which it uses as its intermediate goods, at the price P . The markup creates a wedge between the marginal product of capital and the real interest rate. Based on this, a unit of output requires a composite of capital, labor, and intermediate goods, with intermediate goods proportion λ of firm revenue.

The economy wide markup is:

$$\mu = \frac{(1-\lambda)\gamma}{1-\lambda\gamma},$$

where γ is the firm-level markup and λ is the intermediate-goods share. This markup introduces a wedge between marginal products and real factor prices:

$$\frac{R}{P} = \frac{\partial Y / \partial K}{\mu}.$$

3.1.3 Aggregation and Capital Accumulation

At the aggregate level, we consider the economy in per effective worker terms. The production function is given by:

$$y = f(k) = (1 - \lambda)F(k, 1 - \theta X),$$

where y and k are output and capital per effective worker, respectively, λ is the share of intermediate goods in firm revenue, and θX represents overhead labor scaled by population.

Capital accumulation follows the standard Solow equation:

$$\frac{\partial k}{\partial t} = sf(k) - (n + g + \delta),$$

where s is the constant savings rate, n is the population growth rate, g is the rate of technological progress, and δ is the depreciation rate. In the steady state, capital per effective worker satisfies:

$$sf(k^*) = (n + g + \delta)k^*$$

The return on capital is determined by the rental rate:

$$r + \delta = \frac{f'(k)}{\mu} \Rightarrow r = \frac{f'(k)}{\mu} - \delta.$$

This equation highlights that market power (via $\mu > 1$) drives a wedge between the marginal product of capital and the real interest rate. In the Solow framework, where the savings rate is exogenous and the supply of capital is inelastic, this reduced demand for capital due to market power does not change the steady-state capital stock, but it does lead to a lower equilibrium real interest rate.

3.1.4 Dynamic Efficiency and Pure Profits

We can compare the social return on capital and $n + g$ to be interpreted as dynamically efficient investment. The net marginal product of capital $f' - \delta$ equals the growth rate of the economy $n + g$. Because of diminishing marginal product, if $f' - \delta < n + g$, the economy has more capital than at the Golden Rule steady state, and if $f' - \delta > n + g$, it has less.

The share of value added devoted to pure profits accruing to firms is

$$\Pi = 1 - \frac{1}{\mu} - \varphi X/L ,$$

where φ is the labor share of income WL/PY . Here $1 - \frac{1}{\mu}$ reflects the share of value added that is paid to markups (notice this is zero when $\mu = 1$). Some of the markup goes to overheads, so we subtract off the share going to overheads $\varphi X/L$, leaving the share to pure profits (see also Ball and Mankiw, 2023 p. 576).

Tobin's q links firm valuation to profitability by valuing the firm based on its income less expenditure on capital investment, discounted at rate $r-g$. The value of the firm is then:

$$M = \frac{RK + \Pi Y - (\delta + g)K}{r - g},$$

where we define Tobin's Q as $q = M/K$ (market value over book value of capital). Since $R = r + \delta$, we can then combine the above equations to get:

$$q = 1 + \frac{\Pi y}{(r - g)k} L,$$

which will allow us to solve for Π from parameters for q , k/y , and g and other equations that pin down r . See also Ball and Mankiw (2023, p. 579).

3.2 Operationalizing the Model for Kazakhstan

We now describe how we implement the Ball and Mankiw model using Kazakhstani macroeconomic data. This involves choosing parameter values, linking theoretical expressions to observable statistics, and calculating the implied return on capital and dynamic efficiency conditions.

3.2.1 Baseline Figures

$K/Y = 3$ – The capital stock is three times annual output.

$\delta K = 0.15Y$ – Depreciation is 15% of output, where δ is the depreciation rate.

$1 - \varphi = 0.33$ – Measured gross capital income is 33% of gross output, where φ is the constant labor share of income under the constant returns to scale production function.

$g = 0.02$ – Income per person grows at 2% per year.

$\lambda = 0.45$ – The share of intermediate goods in firm revenue is 45%.

$q = 1.75$ – Firm's MV exceeds the replacement cost of their capital by 75%.

$\gamma = 1.2$ – Firms mark up prices 20% over their marginal cost. Sensitivity analysis allows γ to range from 1.05 to 1.4.

3.2.2 Key Equations Used for Quantification

We use several model equations to map theoretical relationships to Kazakhstani macroeconomic data. The measured return to capital is expressed as:

$$m = \frac{(1-\varphi)}{k/y} - \delta,$$

where m reflects capital income typically reported in national accounts (Ball & Mankiw, 2023, p. 579), δ , is 5% and $1-\varphi$ is the capital share of income. As discussed in Section 2, this would be the capital return under perfect competition with constant returns to scale (after adjusting for depreciation). See also Ball and Mankiw (2023, p. 581).

Pure profits as a share of gross income are computed as:

$$\Pi = (q - 1)(r - g)(k/y),$$

linking Tobin's Q to profitability. The marginal product of capital is calculated as:

$$f'(k) = \mu \left(\frac{m-g}{q} + g + \delta \right),$$

and the corresponding social return on capital is $f'(k) - \delta$. This is compared to $n + g$ to evaluate whether the economy is dynamically efficient.

3.3. Extension to a multi-industry setting

We consider the above model both in aggregate and in a many-industry setting. For the many-industry setting, we allow most parameters to vary by industry, such as the markup, the capital-output ratio, the depreciation rate, and the labor share of income. We allow Tobin's Q to vary by industry when possible. For n and g , we calculate these on an economy-wide basis, as they reflect a typical consumer rather than the income or labor growth of the industry in dynamic efficiency considerations.

3.4. Limitations

A key limitation of our approach is the lack of consideration for cross-industry or cross-country linkages. We consider industries independently, which abstracts from potential for industries to rely on inputs generated from other domestic industries. A second key simplification is the lack of attention to imported intermediates, which was more justifiable in the model's original application to a large economy with a low trade share like the United States. Another potential concern is the absence of more equalization of financial returns to capital across sectors, though this could be justified based on capital market underdevelopment.

A second key limitation is the absence of consideration for externalities related to macroeconomic stability. A crucial concern for Kazakhstan (e.g., as in the NBRK

Strategy 2030, NBRK, 2023), is the lack of economic diversification that leaves the economy exposed to external shocks and creates challenges for monetary policy. Modelling this fully is beyond the scope of this analysis, but at the very least this consideration suggests avoiding activist policy to encourage investment in oil, even if returns are higher.

A third key limitation is the absence of adjustment costs or explicit modeling of challenges in the absorption of capital by sector. What we observe in the data is actual performance which could relate to longer-run fundamentals as well as short-run costs of adjustments. We can infer adjustments costs and within-industry misallocation as possible reasons for poor returns, but distinguishing between these channels is beyond the scope of our analysis.

4. Data and Calculation of Parameters for Calibration

Our main source of data is confidential data from the BNS (2024), which we use to construct the aggregate and industry parameters for calibration. We use two time frames for our analysis. One is to use the latest available data in 2023. A second approach, perhaps more justified in a more volatile emerging market like Kazakhstan, is to estimate parameters over a wider range of data. For the wider range, we choose 2016-2023, with 2016 being a suitable starting point considering the beginning of the NBRK's inflation targeting and the macroeconomic volatility surrounding the end of the fixed exchange rate regime in August 2015.

Table 1 provides the details of how the BNS data was used to calculate parameters relevant for the calibration, suitable to replicate our methods. One difference in our calculation is we have initially calculated a sectoral markup rather than a firm-level markup, though one can easily be calculated from the other under the model

assumptions. For robustness, alternative methods of sectoral markup estimation using frontier econometric methods are presented in Appendix E. The one parameter that does not require BNS data is Tobin's Q. This is the hardest parameter to pin down, as Ball and Mankiw (2023). They use a baseline parameter 1.75 for the U.S. while noting that it varies widely over time. Larger values of this parameter lead to larger estimates of the pure profits that bias upward conventional estimates of the social return to capital. We calculate this using Bloomberg data and discuss details in Appendix C.

We report results of calculated parameters for 2016-2023 in Table 2a and for 2023 in Table 2b, and we discuss our initial attempts to cross-validate aggregate results here. Further validation of our parameters relative to public BNS data is in Appendix D.

For other countries, we complete a similar analysis for Nigeria, Australia, and Norway, which share some features with Kazakhstan. This process is discussed in Appendix B.

5. Results on overinvestment and underinvestment

5.1 Calibrated parameters

Tables 2a and 2b show the results of our calibrated parameters for the aggregate economy and ten 1-digit NACE industries. The results show that our data is capturing substantial variation across industries which have implications for optimal investment. Among standard growth accounting statistics, we observe that utilities has a high capital-labor ratio and depreciation rate, and mining has a high capital share of income.

The 2016-2023 calibrations of Table 2a show basic features of the economy of Kazakhstan, which can be contrasted with calibrations from other countries. The labor share of income 33.7% is much lower than the U.S. value of 67% from Ball and Mankiw (2023). This is largely attributable to the low share for mining in Kazakhstan, equal to 16.1%. The sector markup of 1.719 for Kazakhstan is higher than the U.S. value of 1.43, and mining's sector markup of 2.78 is crucial in driving up Kazakhstan's overall markup. Interestingly, the depreciation rate δ of 10% for Kazakhstan is twice the value calibrated for the U.S. 5%, and 5% is also similar to our estimates in Appendix B for Norway and Australia. Again, the mining plays an important role in Kazakhstan's higher depreciation rate, given its importance for aggregate output and its high depreciation rate of 20%.¹

Our adjustments for perfect competition depend on markups and Tobin's Q. For markups, we observe that mining, trade, and information technology have relatively high markups. These markups could relate to fixed costs (overheads) that reflect returns to capital or excess profits that are distortionary. Higher Tobin's Q for an industry captures the extent of excess profits, according to the Ball and Mankiw (2023) framework laid out in section 3. For Tobin's Q, we estimate an economy-wide value of 1.81 over 6 years and 1.45 for 2023. We have limited ability to estimate across industries due to small sample sizes. Mining is the one exception, for which we estimate a value of 2.96 over 6 years, which we use for both analyses. So we will tend to associate higher markups with higher fixed costs and

¹ For readers familiar with tax depreciation rates, we note that they are widely known to be much higher than economic estimates of depreciation from national statistical data. The idea dates to Samuelson (1964), and Hulten and Wykoff (1978) provide a perspective from the U.S. Treasury on practical challenges in bringing tax treatment of depreciation in line with economic depreciation.

higher marginal of product, while the higher Tobin's Q for mining will adjust downward the returns.

For income growth and population growth, we use the 2016-2023 values for the aggregate economy for all industry analysis and the 2023 analysis, since these are long-run macroeconomic variables used to assess the efficiency of the capital stock. We estimate these to be 5.7%, so we will judge an efficient capital stock to be that achieves a marginal product of capital net of depreciation of 5.7%, for any industry or the aggregate economy.

5.2 Aggregate analysis of overinvestment and underinvestment

The first row of Table 3a and 3b presents our results for the aggregate economy. We find that the net return on capital is 33%, showing that there is substantial room for capital deepening as a source of growth for Kazakhstan. So the excess of the return of the Golden Rule level of 5.7% is 27.3%, suggesting Kazakhstan would still benefit from increased investment on the margin. These results are consistent with the 2023-only results, which suggest a net return of 29% and return in excess of the Golden Rule level equal to 23%.

For the model with perfect competition, the net return on capital is 15% leading to an excess of the Golden Rule of 9.3%. So as in Ball and Mankiw (2023), the adjustment for fixed costs in the modern economy suggests greater returns on capital and benefits of investment than the model of perfect competition. Like the U.S., there is no evidence of dynamic inefficiency in Kazakhstan.

5.3 Industry analysis of overinvestment and underinvestment

Our view of efficiency is one where the capital stock is such that the net marginal product of capital (MPK) is 5.7%, so industries with lower net returns than this are dynamically inefficient and suffering from overinvestment.

We conclude two industries suffer from overinvestment: agriculture with net MPK 4.8% and utilities with net MPK 3.9% for 2016-2023. Returns are similarly low for these two industries in 2023, and we have the additional result of 1% net MPK for manufacturing in 2023. So we conclude that manufacturing has recently suffered from overinvestment. These results occur in a context where investment in utilities is less than half of its 2016-2023 average, while investment in manufacturing and agriculture is higher in 2023 than its 2016-2023 average. These investment outcomes are consistent with recognizing the low returns in utilities and cutting back investment, while pushing ahead with investment in manufacturing and agriculture for the purposes of industrial policy for economic diversification.

We summarize why the data and our model imply overinvestment in agriculture, utilities, and (in 2023 data) manufacturing. In all cases, the high capital-output ratios and low capital share of income suggest low returns to capital, and the relatively low markups imply that adjusting returns upwards due to overheads does not change the conclusion of low returns to capital. We discuss each sector's data in turn, while focusing on the larger data set (2016-2023) for agriculture and utilities.

Utilities (2016-2023). This sector has the highest K/Y ratio (4.685) but a low capital share of income (0.331). Under perfect competition, this data implies an MPK of 1.2%, which is 4.5 percentage points lower than the Golden Rule level of 5.7%. The markup (1.344) is lower for all industries but agriculture and manufacturing. Pure

profits are negative, so this markup is fully devoted to overheads, and the overhead share of gross revenue (12.4%) is similar to the national average. Accounting for these overheads, the MPK increases substantially from 1.2% to 3.9%, but it is still 1.8 percentage points below the Golden Rule level of 5.7%.

Agriculture (2016-2023). This sector has an above-national average K/Y ratio (2.385) and a below average capital share of income (0.308), implying an MPK of 3.1% under perfect competition. We determine pure profits are low (0.5% of gross revenue), so almost all markups are devoted to overheads, but markups (Table 2a) are the lowest in the economy (1.142). The MPK after adjusting for overheads is 4.8%, still 0.9 percentage points below the Golden Rule level of 5.7%.

Manufacturing (2023). Manufacturing has a relatively low capital-output ratio, but the explanation for low returns in 2023 is that the capital share of income fell to an economy-low 0.145, in contrast to a 0.343 average from 2016-2023. Consequently, we find manufacturing had a slightly negative MPK in 2023 under the perfectly competitive model, 6.0 percentage points below the Golden Rule MPK. Pure profits were again low, but the markup also matched agriculture in 2023 for the lowest in the economy (1.109). Overheads were then just 2.5% of gross revenue, so adjusting the MPK for overheads increased the MPK to 1.0%, which is still 4.7 percentage points below the Golden Rule level.

What possibly explains the low MPK in these sectors? Our analysis shows the answer is not market power or barriers to entry, since we find pure profits are low. Also, if market power were a specific problem in these sectors, that would distort upward the perfectly competitive theory's MPK calculations, and these MPKs are also low for these sectors. A fact that stands out about agriculture and utilities is that these industries have a relatively high share of depreciated capital relative

to output ($\delta K/Y$). Here utilities have the highest depreciation of any industry for both 2016-2023 and 2023, and agriculture depreciation is above average. If there are adjustment costs to regularly replacing this capital, perhaps related to government regulation, then this could be a possible explanation for the low MPK in these sectors. This explanation is less relevant for manufacturing, however, where observed depreciation is low.

Our results should be interpreted carefully and not as a rigid endorsement to reduce investment in manufacturing and agriculture in the long run, but rather policy reforms to ensure that current investment is more effective. For these industries, the investment is inefficient, and current policy is not yielding visible results today. One possibility is that these investments will take time to yield returns, which may be plausible for manufacturing, but is certainly not the case for agriculture which has not seen results for 7 years. Possible reasons for poor returns could be adjustment costs in absorbing investment as just discussed, and the pace of investment should be slowed down, or issues in allocating investment within these industries. Because the lower manufacturing returns occur in 2023 as investment into manufacturing accelerated in 2023, either adjustment costs of the new investment or misallocation for the new investment. Such misallocation can be the results of government regulation, such as the need for approval and price regulation. Such results on manufacturing are aligned with other work that has criticized the performance of Kazakhstan's industrial policy for diversification, such as the discussion of Kazakhstan's industrial zones in Lall et al. (2022). Interviews of Kazakhstan's utility firms in heat and power confirm that price regulations preventing increases, for reasons of energy security and social stability, are preventing market-driven firm decisions (Howie & Akhmetov, 2024; Akhmetov & Howie, 2024).

For the industry with the most underinvestment, our results point to information technology. Our analysis of market power is important for these results, because IT is an average industry according to measures for perfect competition, while it is the most underinvested industry once we take into account the importance of fixed costs. We also find that recently, investment in IT has been too low even for its capital stock to continue to increase.

We provide this result with the caveat that our analysis is focused on firms of size 100 and greater given our firm-level data, so we may be missing some investment and high returns for IT firms of size less than 100. Our results then can be interpreted as a need for more investment in successful firms in IT of size greater than 100. This aligns with calls from Tokayev (2023) to encourage the growth of medium-sized firms, as well as the recent World Bank advice (Iacovone et al., 2025) that industrial policy in Central Asia has been too focused on supporting entrepreneurs and small firms and not providing an environment that supports medium firms receiving the capital to grow larger.

Our results also support that the mining sector continues to yield high returns on capital, consistent with mining being the sector that attracts the plurality of investment. This creates challenges for Kazakhstan's economic diversification, but mining is still a substantial source of revenue that could be used for public goods and other support for sectors with underinvestment, so investment in mining need not be discouraged.

5.4 Analysis of experience from other countries.

We present results for Norway, Australia, and Nigeria in Appendix B, which provide some insights into Kazakhstan's experience.

One key insight is that the variation in MPK across the 10 industries is also high in Norway and Australia, and higher in 2023 than in 2016-2023. A first glance of the results for Kazakhstan suggests MPK variation might be due to financial market underdevelopment and that the problem increased in 2023, but similar variation for Norway and Australia suggest that such a result is typical.

Australia is interesting to analyze because we have sufficient data to calculate Tobin's Q across industries, so the Australian results are informative about what we may be missing in our analysis of Kazakhstan with limited industry-specific calculations other than mining.

In Australia, we see a similar result of Kazakhstan where accounting for imperfect competition leads to IT going from a below-average industry to an above-average industry in terms of its marginal product of capital. Interestingly, IT also has a relatively high Tobin's Q, suggesting that some of the industry markups reflect excess profits, but fixed costs are still important. The result gives us confidence that our IT results are not solely a function of underestimating Tobin's Q for IT in Kazakhstan.

5.5 Two-digit NAICS industry analysis

We include in Table G-1 results by two-digit industry. Data is thinner for two-digit industries so we do not estimate the model at the two-digit level, but we can learn more about what two-digit industries are driving results at the one-digit level using data on capital stocks and investment. For agriculture, results are driven by crop and animal production. Forestry and fishing account for a small share of investment. For mining, the bulk of investment and returns are in oil and gas, while other mining activities have returns consistent with the national average.

Low returns in refining appear to be important for the dismal manufacturing performance. The returns in IT appear to be broad-based across subcategories.

5.6 Analysis with production function method of markup estimation

We discuss alternative methods of markup estimation in Appendix E. The production function method of markup estimation may reduce some biases based on the price-cost margin that we use in our earlier analysis. Markups are reduced, but the outcomes do not change the conclusions from our baseline analysis.

6. Results on the Optimal Level of Investment by Sector

We present results for our analysis of the optimal capital stock and annual investment until 2035 across industries in Table 4. We also note that given the scope of our analysis is private firms of size greater than 100, we do not include the public sector or firms under size 100, and this leads us to account for roughly half the capital stock in Kazakhstan.

The model detailed in the previous sections fundamentally provides an answer for the optimal capital stock in the aggregate economy and in each sector, if supplemented with more structure on the nature of decreasing returns.

We have thus far assumed only that the production function is constant returns to scale (CRS) rather than Cobb-Douglas (i.e., a unit elasticity of substitution between capital and labor) , but here we will proceed with this simplifying assumption. Production is then $y = f(k) = Ck^\alpha$ for constant $C = (1 - \lambda)(1 - \theta X/L)^{1-\alpha}$ (see first equation of section 3.1.3). Then $MPK = f' = C\alpha k^{\alpha-1}$ and the decreasing returns to capital $MPK' = f'' = C\alpha(\alpha - 1)k^{\alpha-2} = (\alpha - 1)MPK/k$. Under Cobb-Douglas, we can determine MPK' as a function of MPK , the labor

share of income, and the effective capital stock $k = K/AL$, without needing to calculate C .

To construct an estimate of the necessary change in capital stock to achieve the Golden Rule level, we take a linear approximation of the decreasing returns to capital and determine the amount of capital necessary to bring the MPK down to the Golden Rule level of $n + g + \delta$. Our model for the MPK from the effective capital stock change Δk is then $MPK(\Delta k) = MPK_0 + (\Delta k)(\alpha - 1)MPK_0/k_0$, where MPK_0 and k_0 are the initial MPK and effective capital stock respectively. Now let Δk^* be the effective capital stock change yielding the Golden Rule, such that $MPK(\Delta k^*) = n + g + \delta$. Solving for Δk^* , we obtain

$$\Delta k^* = (1 - \alpha)(1 - \frac{(n+g+\delta)}{MPK_0})k_0.$$

So our approximation for the necessary scale increase in effective capital stock depends on the distance of the MPK from the efficient level, and a measure of the decreasing returns to capital (the labor share of income, $1 - \alpha$). Performing this calculation for Kazakhstan, the optimal capital stock is roughly double the present value (96% for our 2016-2023 model and 92% for our 2023 model).

Our model requires more structure to yield a recommendation for the path of investment to achieve the optimal capital stock. What we can conclude is that higher investment could lead to risks of overinvestment in certain sectors and difficulty in absorbing capital, while lower investment can delay the path toward an efficient capital stock. The overinvestment observed at the sectoral level is small and should not motivate a substantial reduction in overall investment. On this basis, we recommend that investment grows to keep capital accumulating at the same growth rate as the rest of the economy ($g + n$)=5.7%. This is a reasonable

path of investment growth, considering that 2023 investment is approximately 8000 billions in KZT while investment has averaged about 6000 billions KZT from 2016-2023.

On the defined path, the effective capital stock at time T grows linearly in T as $k_T = k_0 + (i - \delta k)T$, where $i = I/AL$ and investment grows so $(i - \delta k)$ is constant over time. Solving for the desired effective capital stock Δk^* we get:

$$T^* = \Delta k^* / (i - \delta k) = (1 - \alpha) \left(1 - \frac{(n+g+\delta)}{MPK_0}\right) k_0 / (i - \delta k).$$

Notice this expression can be written in terms of current capital stock and current investment, rather than effective capital stock and investment, since the present AL terms cancel out. Based on this approach, we obtain an estimate of $T^* = 11.5$ using either 2016-2023 data or 2023 data.

Given this level of investment through 2035, our model can then determine an optimal current level of investment by sector that drives returns in each sector to an efficient level by 2035, assuming net investment grows by $n+g$ (thus, a path where accumulation of effective capital is constant). For these calibrations, we use our industry-specific calibrated parameters, including depreciation rates. To obtain the desired change in capital stock ΔK^* in 12 years, an industry would need effective capital accumulation (investment net depreciation) to satisfy $12(i - \delta k) = \Delta k^*$, so $i = \delta k + \Delta k^* / 12$. As before we can multiply through by AL to avoid needing effective units, so $I = \delta K + \Delta K^* / 12$, where $\Delta K^* = (1 - \alpha) \left(1 - \frac{(n+g+\delta)}{MPK_0}\right) K_0$ is calculated for each industry.

Compared to the status quo, our results in Table 4 suggest shifting more investment to IT and construction, while mining and other services' investment should be above their 2016-2013 average levels but below their 2023 levels. There

is a recommended decrease in investment in utilities similar to what has already occurred, with more moderate decreases in investment in other sectors. With investment allocated more efficiently by sector, the necessary investment to achieve the optimal capital stock is 8% lower.

Investment in construction and IT has already trended upward for 2023 relative to 2016-2023. The model proposes a sharp increase. A note of caution is that the model is consistent with capital accumulation assuming the industry maintains its ability to allocate and absorb capital at current levels of investment, which could be impaired if investment increases too rapidly. Regardless, the current level of investment which is insufficient to allow the capital stock to rise in these two sectors is far too low.

To explain further these results, the key considerations are the current MPK implied by our model in Table 3a and the capital share of income in Table 2a. Higher values of capital share of income imply faster decreasing returns. Although mining and IT have the highest MPK in Table 3a, they have an equilibrium increase in capital stock in Table 4 similar to the rest of the economy (88% for mining, 105% for IT, and 96% for the aggregate economy). Mining and IT then have similar ratios of investment to capital stock; mining ($0.28 = 1829/6604$) has a somewhat higher ratio than IT ($0.22 = 209/964$) given mining's higher depreciation. The increase in recommended IT investment from 85 to 209 billion KZT is much higher, however, given that current IT investment is insufficient even to keep the capital stock constant.

Construction and other services have the highest recommended long run increases in capital stock (241% for construction; 147% for other services), because they have above average MPK in Table 3a, but they also have much lower

capital shares of income than mining or IT, so the implied decreasing returns are lower. While the increased investment for other services from 265 billion KZT to 312 billion KZT is in line with recent history (where the 2023 investment is 630 million KZT), our estimate for optimal construction investment of 274 billion KZT is more than double the 2023 level of 125 billion KZT and triple the longer-term 2016-2023 level of 80 billion KZT.

Our recommended increases in investment and construction are potentially large, but we observe that these are also the industries with among the largest increase of investment in 2023 relative to the 2016-2023 average. Investment was 56% higher in each sector in 2023. The only sector with a higher increase was mining (61% higher), and this is the sector for which we recommended the largest absolute increase in investment, 241 billion KZT. Moreover, our model recommends the largest absolute decreases in investment in utilities and trade, which are also the sectors experiencing the largest recent decreases in 2023 investment relative to the 2016-2023 average. Alignment between the historical changes in investment (which were not a data input in the model) and our recommended changes in investment provides some confirmation of the validity of our methodology and results.

The sectors where our recommended investment change has a distinct direction from recent observed shifts in investment are transport, manufacturing, agriculture, whose 2023 investment relative to 2016-2023 averages are 45%, 29%, and 19% higher, respectively. The recommended increase in capital stock for manufacturing (88%) and transport (87%) is similar but somewhat lower than the rest of the economy (96%). We similarly recommend a somewhat slower pace of investment than the 2016-2023 averages in these two sectors. For agriculture, we

recommend a capital stock decline, given the low calculated MPK. Our model does not capture the motives for recent increases in investment in these three sectors. For transportation, possible motives not captured in our model's calculated private sectoral returns to capital include positive spillovers between transport and other economic sectors. There may also be geopolitical motives tied to the Belt and Road Initiative and development of the Middle Corridor. For manufacturing and agriculture, the model of MPK would not capture any positive spillovers from economic diversification. Similar to our earlier discussion of these two sectors' low returns, the disconnect between observed investment increases and lower recommended investment may also reflect subpar outcomes of state interventions in these two sectors.

We conclude this section noting some further limitations of our approach. Our derivation relied on the Cobb-Douglas assumption. This approach could also be extended to a constant elasticity of substitution production function which is likewise constant results to scale, although these results would require estimates of the elasticity of substitution between capital and labor. We acknowledge that a closed-form solution also exists for the steady-state Golden Rule capital stock when assuming a Cobb-Douglas functional form, but this calculation is highly sensitive to calibration of the capital (or labor) share of income and potential errors in measurement, since they appear as exponents in the closed-form solution. An advantage of Ball and Mankiw (2023) is they focus on calculations that are not exponential functions of any parameters, and our estimates here preserve this feature.

7. Conclusion and Policy Recommendations

We conclude with a discussion of policy recommendations. Our industry analysis has provided us with specific insights into overinvestment and underinvestment by sector. First, our analysis implies that targeted reforms are necessary for agriculture, manufacturing, and utilities to achieve efficiency. These industries have been subject to heavy state involvement and regulation, and the returns are not yet there. Second, our analysis suggests there is scope to attract greater investment in construction and medium-size information technology firms. For construction, resource demands are a potential concern that needs to be balanced with the potential for growth, but returns are high at the margins. There is ample policy supporting entrepreneurship, but there is a lack of financing for medium-sized IT firms to grow to large-sized firms, and our analysis suggests there is scope for encouraging such investment. We conclude there is still a great opportunity for Kazakhstan to grow through capital deepening, and horizontal policies that promote the investment attractiveness of Kazakhstan for domestic and foreign sources remain attractive.

Our analysis offers advice on which sectors should optimally attract increases in investment relative to 2016-2013 averages, based on the returns to capital implied by our model. We recommend the largest increases in investment for mining, construction, IT, and other services. We recommend the largest decreases for utilities and trade. The sign of all these recommendations aligns with recent shifts in investment in these sectors for 2023 relative to their 2016-2023 averages, offering some confirmation for the validity of our analysis. Our recommendations differ though from the observed 2023 upticks in investment in transport, manufacturing, and agriculture. This reveals some limitations of our analysis, since our model's private returns to capital do not capture positive cross-sector spillovers. We acknowledge this limitation is especially important for the

transport sector. While we recognize that our model may not capture macroeconomic benefits of economic diversification in manufacturing and agriculture, the disconnect between increased investment in these two sectors and their lower recommended path of investment in our analysis reaffirms the need to improve the efficiency of state interventions in these two sectors.

Key terms

Overinvestment and dynamic inefficiency – the extent to which capital stocks are so large that investment could be reduced and consumption would increase.

Underinvestment and dynamic efficiency – a state where investments would still increase consumption. The term though does not imply that capital stocks are at the efficient level.

Marginal of product of capital – the derivative of a production function with respect to capital. For constant returns to scale production functions, the MPK is also the first derivative of the production function expressed in per unit of labor, where $y=f(k)$ and y is output per unit of labor and k is per unit of labor.

Markups – the extent to which prices exceed marginal costs.

Fixed costs or overheads – costs that do not change when more output is produced.

Excess profits – profits that go to firm owners rather than lenders of capital, which can only exist in the presence of market power. Markups can go either into fixed costs or excess profits.

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Table 1. Construction of model parameters from raw data

Parameters	Definition
The ratio of capital stock to output, K/Y	(tangible fixed assets + intangible fixed assets) / sales
The ratio of depreciation to output, $\delta K/Y$	amortization / sales
Labor share of income, ϕ	wage bill / (sales – material costs – costs to third parties)
The share of intermediate goods in firm revenue, λ	material costs / sales
Sector markups, μ	$1 / (1 - \text{Lerner index})$, i.e. $1 / (1 - [(sales - material costs - wage bill) / sales])$

Growth in output per worker, g	log difference in sales per worker
Employment growth, n	log difference of total employment

Notes: The parameter notation aligns with that of Ball and Mankiw (2023). All parameters used for the model are constructed from data of the BNS (2024), except for Tobin's q . The definition is sufficient information to reproduce the parameter calculation, once data is available.

Table 2a: Calculated model parameters, 2016-2023, for aggregate economy and NACE1 industries

	Capital stock to output, K/Y	Depreciation $\delta K/Y$	Labor share of income, ϕ	Intermediate share, λ	Sector markup, μ
Aggregate Economy	1.866	0.188	0.337	0.443	1.719
Agriculture	2.385	0.216	0.692	0.713	1.142
Mining	1.355	0.276	0.161	0.27	2.78
Manufacturing	1.304	0.132	0.657	0.732	1.182
Utilities	4.685	0.308	0.669	0.571	1.344
Construction	0.344	0.098	0.816	0.557	1.421
Trade	4.416	0.068	0.284	0.254	2.418
Transport	3.481	0.243	0.392	0.369	1.956
Accommodation	1.253	0.139	0.527	0.45	1.563
Information Technology	1.673	0.216	0.295	0.233	2.375
Other services	1.077	0.105	0.522	0.296	1.736

Notes: Other services include NACE 64-82. Additional parameters for the baseline calibration include growth in output per worker, $g = 0.021$, population growth $n = 0.036$, and Tobin's Q $q=2.96$ for oil and gas and 1.81 for all other sectors.

Table 2b: Calculated model parameters, 2023, for aggregate economy and NACE1 industries

	Capital stock to output, K/Y	Depreciation $\delta K/Y$	Labor share of income, ϕ	Intermediate share, λ	Sector markup, μ
Aggregate Economy	1.989	0.239	0.387	0.475	1.592
Agriculture	2.584	0.334	0.793	0.729	1.109
Mining	1.788	0.398	0.183	0.345	2.26
Manufacturing	1.252	0.163	0.855	0.779	1.109
Utilities	5.646	0.42	0.666	0.54	1.348
Construction	0.319	0.089	0.637	0.526	1.41
Trade	2.869	0.071	0.218	0.135	3.153
Transport	3.712	0.274	0.546	0.418	1.71
Accommodation	1.274	0.163	0.51	0.438	1.529
Information Technology	1.821	0.244	0.323	0.194	2.356
Other services	1.228	0.116	0.552	0.331	1.59

Notes: Other services include NACE 64-82. Additional parameters for the baseline calibration include growth in output per worker, $g = 0.021$, population growth $n = 0.036$, and Tobin's Q $q=2.96$ for oil and gas and 1.45 for all other sectors.

Table 3a: Model results, using parameters for 2016-2023

	Net return on capital, $f-\delta$	Return on financial assets that accumulate capital, r	Share of pure profits in gross revenue, $\Pi(1-\lambda)$	Overhead share of gross revenue	Net return on capital less growth, $f-\delta-n-g$	Net return on capital less growth, $f-\delta-n-g$ (ignoring market power)
Aggregate Economy	0.330	0.150	0.109	0.124	0.273	0.093
Agriculture	0.048	0.031	0.005	0.030	-0.009	-0.026
Mining	0.792	0.154	0.258	0.209	0.735	0.097
Manufacturing	0.135	0.099	0.022	0.019	0.078	0.042
Utilities	0.039	0.012	-0.015	0.124	-0.018	-0.045
Construction	0.329	0.147	0.016	0.116	0.272	0.090
Trade	0.240	0.090	0.186	0.252	0.183	0.033
Transport	0.198	0.067	0.083	0.226	0.141	0.010
Accommodation	0.307	0.156	0.076	0.122	0.250	0.099
Information Technology	0.583	0.171	0.156	0.288	0.526	0.114
Other services	0.420	0.201	0.110	0.188	0.363	<i>0.144</i>

Notes: Other services include NACE 64-82. Results are based on sector parameters in Table 2a. Bolded parameters are negative, indicating the model suggests overinvestment. The italicized parameters are for the industry with the greatest extent of underinvestment (based on returns)

Table 3b: Model results, using parameters for 2023

	Social return on capital, $f'-\delta$	Return on financial assets that accumulate capital, r	Share of pure profits in gross revenue, $\Pi(1-\lambda)$	Overhead share of gross revenue	social return on capital less growth, $f'-\delta-n-g$	social return on capital less growth, $f'-\delta-n-g$ (ignoring market power)
Aggregate Economy	0.288	0.136	0.054	0.141	0.231	0.079
Agriculture	-0.016	-0.027	-0.015	0.042	-0.073	-0.084
Mining	0.491	0.093	0.165	0.200	0.434	0.036
Manufacturing	0.010	-0.003	-0.003	0.025	-0.047	-0.060
Utilities	0.021	-0.004	-0.029	0.148	-0.036	-0.061
Construction	0.959	0.599	0.039	0.099	<i>0.902</i>	<i>0.542</i>
Trade	0.613	0.177	0.175	0.416	0.556	0.120
Transport	0.121	0.040	0.018	0.223	0.064	-0.017
Accommodation	0.348	0.184	0.052	0.142	0.291	0.127
Information Technology	0.583	0.171	0.099	0.365	0.526	0.114
Other services	0.363	0.193	0.064	0.185	0.306	0.136

Notes: Other services include NACE 64-82. Results are based on sector parameters in Table 2b. Bolded parameters are negative, indicating the model suggests overinvestment. The italicized parameters are for the industry with the greatest extent of underinvestment (based on returns).

Table 4. Model for optimal investment levels until 2035

All level variables in billions of KZ tenge

	Capital stock	% increase in capital stock necessary to achieve Golden Rule	Average investment, 2016-2023	Investment 2023	Optimal annual investment	Change from 2016-2023 average
Agriculture	701	-21.3%	158	188	51	-107
Mining	6604	88.0%	1589	2565	1829	241
Manufacturing	5462	96.2%	1069	1382	991	-78
Utilities	3754	-52.2%	646	308	84	-563
Construction	565	240.9%	80	125	274	194
Trade	4575	100.1%	690	128	452	-238
Transport	8377	86.7%	1333	1935	1190	-143
Accommodation	246	126.4%	53	-45	53	0
Information Technology	964	104.8%	85	133	209	124
Other services	1418	146.8%	265	630	312	47

Note: 2035 is selected as the year that aggregate capital stock reaches the optimal return at current investment and depreciation rates.

Appendix B tables

Table B1a Australia parameters, 2016-2023

	Capital stock to output, K/Y	Depreciation $\delta K/Y$	Labor share of income, ϕ	Intermediate share, λ	Sector markup, μ	Tobin's Q , q
Aggregate Economy	1.809	0.099	0.592	0.421	1.671	1.22
Agriculture	1.563	0.142	0.366	0.492	1.727	1.17
Mining	3.216	0.184	0.170	0.169	3.932	1.38
Manufacturing	0.452	0.047	0.602	0.583	1.373	1.47
Utilities	0.160	0.018	0.556	0.402	1.786	0.81
Construction	0.125	0.011	0.668	0.779	1.175	1.69
Trade	0.171	0.013	0.605	0.712	1.225	1.69
Transport	2.811	0.155	0.538	0.168	2.629	1.56
Accommodation	0.765	0.043	0.690	0.337	1.666	2.72
Information Technology	2.175	0.139	0.482	0.174	2.747	2.22

Notes: Additional parameters for the baseline calibration include growth in output per worker, $g = 0.038$, population growth $n = 0.023$.

Table B1b, Australia results, 2016-2023

	Social return on capital, $f-\delta$	Return on financial assets that accumulate capital, r	Share of pure profits in gross revenue, $\Pi(1-\lambda)$	Overhead share of gross revenue	social return on capital less growth, $f-\delta-n-g$	social return on capital less growth, $f-\delta-n-g$ (ignoring market power)
Agg Economy	0.282	0.147	0.025	0.207	0.221	0.086
Agriculture	0.539	0.274	0.032	0.182	0.478	0.213
Mining	0.783	0.156	0.119	0.501	0.722	0.096
Manufacturing	0.779	0.539	0.045	0.069	0.718	0.478
Construction	5.971	3.293	-0.060	0.323	5.910	3.233
Wholesale Trade	1.813	1.530	0.029	0.004	1.753	1.470
Retail Trade	1.651	1.333	0.044	0.009	1.590	1.273
Transport	0.310	0.084	0.060	0.456	0.249	0.023
Accommodation	0.291	0.152	0.100	0.165	0.230	0.092
Information Technology	0.385	0.100	0.134	0.391	0.324	0.039

Note: Results are based on sector parameters in Table B1a.

Table B2a, Norway parameters 2016-2023

	Capital stock to output, K/Y	Depreciation $\delta K/Y$	Labor share of income, φ	Intermediate share, λ	Sector markup, μ
Agg Economy	1.904	0.106	0.618	0.453	1.382
Agriculture	3.381	0.231	0.483	0.572	1.388
Mining	0.958	0.111	0.753	0.620	1.218
Manufacturing	0.466	0.052	0.801	0.731	1.099
Construction	0.341	0.027	0.746	0.667	1.113
Trade	0.379	0.045	0.732	0.444	1.222
Transport	0.423	0.085	1.025	0.678	1.038
Accommodation	0.523	0.040	0.823	0.540	1.126
Information Technology	0.569	0.105	0.770	0.502	1.233
Other services	4.938	0.157	0.527	0.404	1.582

Notes: Other services include NACE 64-82. Additional parameters for the baseline calibration include growth in output per worker, $g = 0.048$, population growth $n = 0.011$, and Tobin's $Q=1.38$

Table B2b, Norway results, 2016-2023

	Social return on capital, $f-\delta$	Return on financial assets that accumulate capital, r	Share of pure profits in gross revenue, $\Pi(1-\lambda)$	Overhead share of gross revenue	social return on capital less growth, $f-\delta-n-g$	social return on capital less growth, $f-\delta-n-g$ (ignoring market power)
Agg Economy	0.184	0.118	0.028	0.123	0.125	0.059
Agriculture	0.130	0.075	0.015	0.105	0.070	0.015
Mining	0.166	0.116	0.009	0.059	0.107	0.056
Manufacturing	0.277	0.242	0.009	0.015	0.218	0.183
Construction	0.561	0.496	0.019	0.014	0.502	0.437
Trade	0.562	0.438	0.031	0.070	0.503	0.379
Transport	-0.175	-0.176	-0.012	0.023	-0.234	-0.235
Accommodation	0.238	0.203	0.014	0.037	0.179	0.144
Information Technology	0.255	0.172	0.013	0.081	0.195	0.112
Other services	0.113	0.060	0.013	0.206	0.053	0.000

Notes: Other services include NACE 64-82. Results are based on sector parameters in Table B2a.

Table B3a, Nigeria parameters, 2016-2023

	Capital stock to output, K/Y	Depreciation $\delta K/Y$	Labor share of income, ϕ	Intermediate share, λ	Sector markup, μ
Agg Economy	0.320	0.057	0.493	0.217	1.868
Agriculture	1.642	0.062	0.237	0.242	3.143
Mining	3.592	0.044	0.279	0.651	1.475
Manufacturing	0.678	0.118	0.040	0.249	3.659

Notes: Other services include NACE 64-82. Additional parameters for the baseline calibration include growth in output per worker, $g = 0.039$, population growth $n = 0.041$, and Tobin's $Q=1.07$

Table B3b, Nigeria Results, 2016-2023

	Social return on capital, $f'-\delta$	Return on financial assets that accumulate capital, r	Share of pure profits in gross revenue, $\Pi(1-\lambda)$	Overhead share of gross revenue	social return on capital less growth, $f'-\delta-n-g$	social return on capital less growth, $f'-\delta-n-g$ (ignoring market power)
Agg Economy	2.605	1.312	0.024	0.340	2.525	1.232
Agriculture	1.339	0.400	0.033	0.483	1.259	0.320
Mining	0.269	0.178	0.013	0.100	0.189	0.098
Manufacturing	4.722	1.164	0.040	0.505	4.642	1.084

Notes: Other services include NACE 64-82. Results are based on sector parameters in Table B3a.

Table G1. Capital Stocks and investment by Kazakhstan two-digit industry, 2016-2023

NACE2	Capital stock (Billion KZT)	Capital labor ratio	Sales to labor ratio	Investment (Billion KZT)
1. Crop and animal production	698.377	12.864	5.354	156.735
2. Forestry and logging	1.667	1.16	1.705	0.441
3. Fishing and aquaculture	0.683	1.663	2.318	0.09
5. Mining of coal and lignite	120.33	9.22	6.678	27.616
6. Extraction of crude petroleum and natural gas	4741.357	111.024	85.05	1076.262
7. Mining of metal ores	1523.739	36.177	21.725	484.465
8. Other mining and quarrying	63.318	8.785	5.605	13.868
9. Mining support service activities	155.391	6.601	10.805	6.8
10. Manufacture of food products	339.03	7.797	11.377	97.654
11. Manufacture of beverages	158.143	20.715	14.639	21.374

12. Manufacture of tobacco products	30.99	26.745	101.98 3	2.458
13. Manufacture of textiles	13.9	4.412	5.791	2.226
14. Manufacture of wearing apparel	8.207	1.802	2.649	2.009
15. Manufacture of leather and related products	1.764	2.236	3.614	0.06
16. Manufacture of wood and products of wood and cork, except furniture	1.052	4.134	2.107	0.209
17. Manufacture of paper and paper products	29.83	11.227	9.388	9.584
18. Printing and reproduction of recorded media	18.546	15.016	9.976	-1.734
19. Manufacture of coke and refined petroleum products	1287.95 8	153.14 7	22.181	243.515

20. Manufacture of chemicals and chemical products	242.44	19.608	14.17	47.694
21. Manufacture of basic pharmaceutical products	49.283	16.221	15.287	7.466
22. Manufacture of rubber and plastics products	29.817	8.711	13.267	11.262
23. Manufacture of other non-metallic mineral products	375.892	16.071	9.775	82.804
24. Manufacture of basic metals	2437.696	31.661	24.964	439.103
25. Manufacture of fabricated metal products, except machinery and equipment	37.942	5.048	8.018	13.305

26. Manufacture of computer, electronic and optical products	5.043	9.193	22.327	0.167
27. Manufacture of electrical equipment	43.506	7.063	14.266	14.318
28. Manufacture of machinery and equipment n.e.c.	71.306	5.55	8.344	18.151
29. Manufacture of motor vehicles, trailers and semi-trailers	56.021	18.85	67.865	17.084
30. Manufacture of other transport equipment	73.014	17.378	24.055	-2.066
31. Manufacture of furniture	3.905	3.724	6.991	-0.331
32. Other manufacturing	10.445	12.029	9.418	1.352
33. Repair and installation of machinery and equipment	136.55	6.401	8.842	-2.965

35. Electricity, gas, steam and air conditioning supply	2958.404	32.686	7.799	409.523
36. Water collection, treatment and supply	736.632	28.95	2.42	208.728
37. Sewerage	42.887	52.409	2.239	31.568
38. Waste collection, treatment and disposal; materials recovery	26.817	3.622	3.975	9.61
39. Remediation activities and other waste management services	0.112	0.851	1.88	0.046
41. Construction of buildings	147.785	3.882	14.021	28.199
42. Civil engineering	291.539	6.12	15.899	20.023
43. Specialized construction activities	125.693	4.258	12.029	12.967

45. Wholesale and retail trade; repair of motor vehicles and motorcycles	41.099	7.352	8.057	15.302
46. Wholesale trade, except of motor vehicles and motorcycles	4266.56	88.855	15.988	468.014
47. Retail trade, except of motor vehicles and motorcycles	267.1	3.963	4.73	79.228
49. Land transport and transport via pipelines	7159.76 4	178.03	44.506	906.178
50. Water transport	28.73	91.021	71.742	-1.985
51. Air transport	408.324	53.9	30.982	154.757
52. Warehousing and support activities for transportation	742.875	25.669	11.56	208.37
53. Postal and courier activities	37.392	31.814	25.889	1.622

55. Accommodation	201.444	25.682	10.203	46.426
56. Food and beverage service activities	44.339	1.831	4.797	8.98
58. Publishing activities	2.649	2.142	3.289	0.61
59. Motion picture, video and television program production	9.466	12.087	6.556	-0.719
60. Programming and broadcasting activities	65.269	22.066	12.054	10.613
61. Telecommunications	824.533	94.204	50.283	43.126
62. Computer programming, consultancy and related activities	38.011	6.783	8.244	20.119
63. Information service activities	24.551	3.406	6.185	2.377
64. Financial service activities, except insurance and pension funding	19.827	10.244	51.592	9.414

66. Activities auxiliary to financial services and insurance	4.788	6.082	13.753	1.532
68. Real estate activities	285.225	21.889	6.622	58.847
69. Legal and accounting activities	1.613	0.748	6.011	-0.594
70. Activities of head offices; management consultancy activities	105.582	20.203	68.766	53.467
71. Architectural and engineering activities; technical testing	235.979	14.933	14.808	-31.114
72. Scientific research and development	104.509	14.119	8.211	-4.671
73. Advertising and market research	0.375	0.459	4.463	-0.017

74. Other professional, scientific and technical activities	75.665	11.895	9.667	-10.889
75. Veterinary activities	14.375	3.604	2.3	-1.83
77. Rental and leasing activities	44.948	17.355	17.631	13.963
78. Employment activities	4.912	0.201	5.754	1.771
79. Travel agency, tour operator and other reservation service activities	2.231	2.648	5.427	-3.348
80. Security and investigation activities	12.215	0.477	2.822	2.775
81. Services to buildings and landscape activities	42.73	3.016	3.181	18.762
82. Office administrative and support activities	462.655	124.626	24.7	121.226

Appendix A. Literature review

A.1 The optimal level of investment required to achieve long-term sustainable economic growth

Certain endogenous growth models highlight how innovation and technical advancements derived from R&D investments contribute to long-term economic growth. Using the variables of R&D investment of various qualities, such as R&D investments in the public sector, private sector, or universities, total R&D investments, and staff numbers in R&D field, Yazgan and Yalcinkaya (2018) add to the economic literature in this regard by analyzing the effects of R&D investments on economic growth while taking into account the income level differences of OECD (OECD-20 and OECD-9) member countries. Their findings indicate that investments in R&D, as well as the accumulation of human and physical capital, have a major influence on economic growth. To lessen income gaps with the OECD-20 group, OECD-9 nations must prioritize investing in human capital, strengthening sector cooperation, and boosting the quality of R&D.

It's interesting to note that the literature has placed a lot of emphasis on the function of agriculture in the early phases of economic growth, emphasizing how it helps with structural change and poverty alleviation. In order to achieve long-term growth, which is explained by labor leave from agricultural sectors to non-agricultural sectors, Gollin et al. (2002) state that agricultural growth and productivity (due to modern technology and development) are important in the early stages based on the structural transformation model (basic neoclassical model with an explicit agricultural sector variable). They argue that structural transformation, which happens when workers move from agriculture to more productive industries, is one of the primary drivers of total productivity development. Accordingly, Badibanga and Ulimwengu (2020) put up a two-sector model of economic growth that aims to ascertain the ideal amount of agricultural investment required for the Democratic Republic of Congo (DRC) to have sustained growth and reduce poverty. Their results indicate that between 2002 and 2012, agricultural investment in the DRC was just 45% of its optimal level. They contend that boosting agricultural investment to its ideal level would accelerate the rise of the agricultural GDP and reduce poverty. Nevertheless, they warn that even reaching the optimal level of investment would not be enough to completely satisfy development goals on its own, especially given other underlying constraints including inadequate infrastructure and governance problems, notwithstanding the possible advantages of optimal investment. McMillan et al. (2014) take a more thorough approach by looking at how structural change affects productivity

growth, particularly in Africa and Latin America. They argue that while many African countries have seen a labor force transition from low-productivity to high-productivity industries, the process has been sluggish and, in some cases, has even slowed growth, particularly in Latin America. This suggests that while increased foreign direct investment and globalization have opened up new economic opportunities, structural change is not always unavoidable, especially when it comes to governance-related concerns. These findings are in line with Gollin's study, which demonstrates how agricultural growth can serve as the foundation for more substantial economic shifts. Similarly, Saidi et al. (2023) demonstrate through a panel threshold model that poverty reduction requires improved governance, particularly in nations that heavily rely on agriculture and have governance issues.

Brock and Mirman (1972) bring uncertainty into optimal growth models by adding a random component to the production function. Their model shows that even in the presence of random fluctuations, efficient investment and consumption policies can remain stable to a level comparable to deterministic models. The influence of output price uncertainty on investment is examined by Abel (1983). He demonstrates that regardless of the form of the marginal adjustment cost function, uncertainty typically results in increased investment. Uncertainty causes investment to increase more slowly than anticipated if the marginal adjustment cost is convex, which means it gets more expensive to invest more. On the other hand, uncertainty causes investment to rise more quickly if the marginal adjustment cost is concave, which means that it gets cheaper to invest more as investment grows. According to Abel's model, a greater degree of uncertainty boosts the projected value of future marginal revenue products of capital, which in turn leads to higher overall investment.

In their analysis of the growth of market power in the US economy between 1980 and 2016, De Loecker et al. (2020) discovered that aggregate markups rose dramatically from 21% above marginal cost in 1980 to 61% in 2016. A few powerful companies are largely responsible for this increase, which is causing market concentration and increased inequality. These companies have become more profitable, but the median markup has stayed the same, indicating that larger companies control a disproportionate amount of revenue and dominate their respective industries. Because dominant corporations inhibit competition and innovation, the study ties the increase in market power to lower labor share and less economic mobility. The results highlight the necessity of reviewing antitrust laws in order to stop monopolistic behavior and preserve competition.

A.2 Policy institutions perspective

Gelb (2010) contends that resource-rich nations ought to use horizontal strategies to give human capital development and macroeconomic stability top priority. Trade openness and educational spending are two examples of measures that lessen the risks of economic volatility associated with a high reliance on natural resources. This is in line with the conclusions of Deléchat et al. (2024), who emphasize the importance of comprehensive reforms including fixing market flaws and enhancing infrastructure. Both sources, however, warn that in order to prevent market distortions and inefficiencies that could worsen the resource curse, vertical policies - which focus on particular industries, such infrastructure or export diversification - must be properly crafted.

This idea is echoed by Pouokam (2023), who highlights the need for investments in human capital, innovation, and technological diffusion in order to overcome the resource curse. She uses Norway and Botswana as examples of how smart fiscal policies and good governance, including the Permanent Income Hypothesis (PIH), have allowed these nations to leverage their resource richness for long-term, sustainable growth. This is consistent with Berg et al. (2013), who support a sustainable investing strategy that strikes a balance between public investment and the creation of a Sovereign Wealth Fund (SWF) in order to stabilize economies and prevent the impulsive spending of natural resource earnings. This idea is further supported by Gelb, Tordo, and Halland (2014), who contend that in order to maintain long-term stability and economic diversity, SWFs should concentrate on domestic investments, especially in infrastructure.

Izvorski et al. (2018) add a crucial point by pointing out the pro-cyclical fiscal policies that are common in many resource-rich nations, notwithstanding these unanimity on the importance of strategic investment. Economic stability is weakened by these policies, which cause fluctuating government spending during boom and bust cycles. In order to manage resource riches more predictably and enable governments to sustain infrastructure development and investment levels during downturns, Izvorski advocates for counter-cyclical fiscal measures. This suggestion is in line with Deléchat et al. (2024) and Pouokam (2023), who also stress the necessity of institutional capacity building and governance reforms to promote economic diversification and resilience against changes in the price of commodities globally.

A.3 Analysis of overinvestment, underinvestment, and optimal investment with recommendations

The problem of inefficient investment is a recurrent theme in the literature and is frequently associated with agency issues. To comprehend the problems of overinvestment and underinvestment, agency theory - which looks at the conflict of interest between managers and shareholders - is essential. According to Jensen (1986), overinvestment usually happens when managers pursue their own goals (such as empire-building) by making investments that expand the company's size without increasing its value. Conversely, underinvestment is the failure of businesses to take advantage of worthwhile investment possibilities because of budgetary limitations, especially in businesses with significant debt or little room for expansion (Myers, 1977).

La Rocca et al. (2005) emphasizes how disputes among managers, shareholders, and debtholders affect capital structure choices and result in less-than-ideal investments. The issue of overinvestment, in which managers put their own interests ahead of the company's long-term profitability, is frequently made worse by the division of ownership and management (Jensen & Meckling, 1976). On the other hand, because managers and owners have a stronger alignment of interests, the incentives for overinvestment and underinvestment may be different in companies with concentrated ownership, such as family-owned enterprises (Zhang, 2009).

Furthermore, because of liquidity constraints, financially pressured firms may struggle with underinvestment during recessions but are more likely to overinvest following favorable cash-flow shocks (He & Kondor, 2016). A major inefficiency seen in many markets is the cyclical pattern of underinvestment during downturns and overinvestment during economic booms. Well-thought-out interventions are necessary to smooth out these cycles because policies intended to reduce underinvestment during recessions frequently make overinvestment worse during booms (He & Kondor, 2016).

Ho et al. (2017) used the Red Queen competition paradigm, which provides an intriguing lens through which to view how businesses modify their investments in relation to industry norms. According to the study, a company's performance may be impacted if it deviates from the industry norm, whether through excessive or insufficient investment. Businesses may overinvest in an effort to beat rivals by making larger investments than the industry average, however this results in decreasing returns. On the other hand, businesses in sectors where technical innovation is essential to preserving competitiveness may suffer from underinvestment. Although it is still centered on IT

rather than more general capital investments, this model emphasizes how market dynamics and competition influence investment behavior and outcomes.

Policy interventions have been proposed to optimize investment, given the prevalence of both overinvestment and underinvestment. Several academics, for instance, contend that enhancing corporate governance procedures - such as matching managerial incentives with long-term company performance - can reduce overinvestment (La Rocca et al., 2005). Managers are less likely to pursue projects that enrich them personally but do not optimize firm value when incentive-based mechanisms, including executive salary tied to shareholder value, are in place.

Financial support tools like loan guarantees and subsidized credit can help ease liquidity issues and promote investment when there is underinvestment, especially in industries like agriculture that are frequently neglected by conventional finance methods (Mishra, 2024). For instance, the Agricultural loan Guarantee Scheme (ACGS) and other agricultural finance initiatives in Nepal have been effective in expanding loan availability and promoting investment in cutting-edge agricultural technologies (Mishra, 2024). To properly address underinvestment in developing industries, more focused actions are required, as evidenced by the ongoing problems of high interest rates and gaps in financial literacy.

He & Kondor (2016) offer additional macroeconomic-level proposals, such as the implementation of price support policies that stabilize capital prices during recessions, potentially mitigating investment boom-bust cycles. According to their model, if carefully crafted, such regulations could lower investment volatility by modifying firms' incentives to make investments during difficult economic times.

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Appendix B. Country data points for Ball-Mankiw 2023 model calibration

B.1 Australia

The Australian Bureau of Statistics (ABS) provides comprehensive data covering both the aggregate economy and individual industries, which is essential for calibrating the Ball-Mankiw (2023) model. This dataset includes key macroeconomic and industry-specific variables. The data obtained from the ABS encompasses parameters such as sales revenue, material costs, wages and salaries, depreciation, capital stock, persons employed, and expenses to third parties.

By utilizing data from the Australian Bureau of Statistics, all required parameters for calibrating the Ball-Mankiw model are sourced from a reliable national authority. This comprehensive dataset supports a robust analysis both at the aggregate and industry levels, enhancing the accuracy of model calibration for the Australian economy.

The Income Statement of Industry from the Australian Bureau of Statistics (ABS) provides detailed financial information across various sectors of the Australian economy. The data obtained from this source includes key financial metrics such as total sales revenue, material costs, wages and salaries, and expenses paid to third parties. These indicators offer insights into the revenue generation and cost structure of different industries, allowing for an analysis of operational efficiency and financial performance. The dataset is particularly useful for understanding how industries allocate resources and manage their operating costs over time. This information supports comparative analysis across sectors and helps to identify trends in production costs and labor expenditures within the Australian economy.

The Annual Estimates of Key Economic Measures from the Australian Bureau of Statistics provides comprehensive data on various macroeconomic indicators, including depreciation and capital stock and persons employed. The depreciation data reflects the consumption of fixed capital, representing the reduction in the value of physical assets over time due to wear and tear or obsolescence. This measure is crucial for understanding the rate at which productive assets lose value across different industries. The capital stock data captures the total value of physical assets used in the production process, including machinery, buildings, and infrastructure. This dataset provides insights into the investment patterns and productive capacity of the Australian economy. Together, these indicators help track the sustainability of capital resources and the long-term growth potential of various sectors.

B.2 Norway

Like Australia, Statistics Norway provides detailed data for both the aggregate economy and individual industries, which is essential for calibrating the Ball-Mankiw (2023) model. The data includes sales revenue, material costs, wages, depreciation, capital stock, third-party expenses, and employment figures. At the national level, the data captures overall economic performance, helping to analyze price rigidity across the economy. At the industry level, it provides insights into sector-specific cost structures and price adjustments.

The Production Account and Income Generation by Industry dataset from Statistics Norway provides detailed financial and economic information across various sectors of the Norwegian economy. The data obtained includes sales revenue, material costs, depreciation, and expenses to third parties. Sales revenue reflects the total income generated from goods and services, while material costs represent the expenditure on raw materials and other inputs required for production. Depreciation data captures the reduction in the value of physical assets over time, indicating the rate at which capital stock is consumed. Expenses to third parties cover payments made to external entities for services or supplies. This dataset is valuable for analyzing industry-level performance, cost structures, and resource allocation within Norway's economy.

B.3 Nigeria

Unlike Australia and Norway, Industry-level data were obtained from annual reports of listed firms on the Nigerian Stock Exchange, based on the financial statements available during the calibration period. These reports provided detailed information on sales revenue, material costs, depreciation, and labor expenses specific to various sectors. Aggregate economic data was sourced from the national statistical authority.

The data for Nigeria was sourced from the National Bureau of Statistics (NBS), covering macroeconomic indicators, inflation rates, and sectoral growth trends and also the financial statements of firms listed on the Nigerian Stock Exchange. This data is useful for calibrating the Ball-Mankiw model in an emerging market context.

B.4 Malaysia

For Malaysia, obtaining aggregate economic data proved challenging due to missing parameters on the national statistical platform. Key variables required for calibrating the Ball-Mankiw (2023) model, such as sales revenue, material costs, wages, and capital stock, were incomplete or unavailable.

Despite these data gaps, it remains possible to calibrate the Ball-Mankiw model, although doing so will require additional time to explore alternative data sources. These may include international financial databases, industry reports, and corporate financial statements. Accessing a broader range of data repositories is essential to ensure a more accurate calibration of the model, reflecting Malaysia's economic conditions and sectoral dynamics.

Malaysia's data was collected from Bank Negara Malaysia and the Department of Statistics Malaysia (DOSM). The dataset includes key monetary indicators, exchange rates, and industry-specific performance measures.

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Appendix C. Tobin's Q calculations and comparisons to prior work

This Appendix section details our calculations of Tobin's Q and compares results to prior work. For all four countries, Tobin's Q was calculated on an annual basis by summing the market value of firm assets (equity and debt) across firms and dividing by book value of assets of all firms, according to firm-level financial data from Bloomberg. Data is labeled based on the beginning of the year indicated, i.e. last date of the previous year. The average is the simple average from 2017-2023.

Tobin's Q values may fluctuate due to market volatility. We compare with previously published estimates of Tobin's Q by country. Our results are broadly consistent with Kazakhstan having higher values of Q and Nigeria having the lowest values of Q.

Table C.1 Tobin's Q figures and benchmarks

Country	N	2017-2023	2023	Benchmark
Kazakhstan	63	1.81	1.45	1.68 (for 2010-2018, N=150, from Sakenova & Arslan, 2022)
Norway	309	1.38	1.23	1.22 (for 1999-2004, N=139, from Chua et al., 2010)
Australia	1790	1.22	1.15	1.65 (for 1999-2004, N=892, from Chua et al., 2010)
Nigeria	151	1.11	1.07	0.93 (for 2003-2007, N=101, From Ibhagui et al., 2018)

Source: Author's calculations using Bloomberg data

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Appendix D. Comparison of estimated Kazakhstan parameters to public data.

Table D1 provides, where available, comparisons of public data to the parameters estimated from our confidential data. We reiterate that confidential data is necessary for a complete set of parameters, especially at the 1-digit and 2-digit NACE levels. We note that some deviation between the two sources is possible since our confidential data does not consider data from firms with employment less than 100, while national statistics use surveys of these small firms. We see that the data largely matches well with the public sources.

Table D.1. Comparison of aggregate data with public BNS data.

Parameters	Aggregated firm-level BNS data	Public BNS data
The ratio of capital stock to output, K/Y	1.866	1.804
The ratio of depreciation to output, $\delta K/Y$	0.188	0.219
Labor share of income, ϕ	0.337	0.307
The share of intermediate goods in firm revenue, λ	0.443	0.423
Sector markups, μ	1.719	1.734
Growth in output per worker, g	0.021	0.035
Employment growth, n	0.036	0.012

Appendix E. Alternative methods of markup estimation.

E.1. Issues with the Accounting PCM Method

The traditional approach to estimating firm-level markups relies on the price-cost margin (PCM), calculated as the ratio of price to marginal cost. In practice, marginal costs are typically unobservable, leading researchers to approximate them using average costs. However, this method has significant limitations. It assumes perfect competition and constant returns to scale, which may not hold in real-world settings. Additionally, it does not account for firm-specific productivity shocks, demand variations, or dynamic input adjustments.

To overcome these issues, we adopt the structural estimation approach of De Loecker and Warzynski (2012), which derives markups from cost minimization conditions using output elasticities from estimated production functions. We implement this method using the control function approach proposed by Akerberg, Caves, and Frazer (2015) to address simultaneity bias in production function estimation.

E2. Overview of DLW (2012) and ACF (2015) Methods

The DLW (2012) approach estimates markups using firm-level cost minimization behavior. Under the assumption that firms minimize costs given their production technology, the first-order condition for a variable input yields an expression for markup as the ratio of output elasticity to the expenditure share of that input. This method does not require direct observation of marginal costs, making it a flexible and widely used alternative to accounting-based approaches.

Production function estimation is central to this methodology, as output elasticities are key to computing markups. However, standard OLS estimation of production functions suffers from simultaneity bias—firms observe their productivity when choosing input levels, leading to correlation between inputs and unobserved productivity shocks. Akerberg, Caves, and Frazer (2015) propose a solution by using a control function approach that relies on investment or materials as proxies for productivity, removing endogeneity concerns.

E3. Estimating the Production Function

We estimate a Cobb-Douglas production function where output is determined by capital and a composite input that combines labor and materials costs, approximating the cost of goods sold. The production function is specified as:

$$Q_{it} = A_{it} K_{it}^{\beta_K} X_{it}^{\beta_X}$$

where Q_{it} is output, K_{it} is capital, and X_{it} is the composite input. The term A_{it} represents firm-specific productivity, which is unobservable but can be controlled for using the ACF method.

The estimation follows a two-stage procedure. In the first stage, we use a nonparametric projection to remove measurement errors and unexpected output shocks, regressing output on inputs and a control variable (investment or materials). This helps account for firm-level productivity that is correlated with input choices. In the second stage, we estimate the production function input parameters using GMM.

E4. Markup estimates

Once output elasticities are obtained, we compute markups as:

$$\mu_{it} = s_{it} / \beta_X$$

where s_{it} is the expenditure share of the composite input in total revenue. This structural approach provides firm-level markup estimates that account for differences in production technology and firm behavior.

The table below compares the markup estimates obtained using different methods.

Table E1. Firm-Level Markup Estimates: PCM vs. DLW (2012) Approach

	markups, pcm	markups, production function estimation	markups, pcm	markups, production function estimation
	2016-2023		2023	
Aggregate Economy	1.719	1.379	1.592	1.35
NACE1				
Agriculture	1.142	0.98	1.109	0.99
Mining	2.78	1.11	2.26	1.34
Manufacturing	1.182	1.18	1.109	1.18
Utilities	1.344	1.29	1.348	1.22
Construction	1.421	1.22	1.41	1.29
Trade	2.418	1.88	3.153	2.08

Transport	1.956	1.42	1.71	1.33
Accommodation	1.563	1.36	1.529	1.3
Information Technology	2.375	1.9	2.356	2.04
Other services	1.736	1.49	1.59	1.6

As shown in Table E1, markup estimates using production functions are generally lower, which is standard and consistent with the literature. However, overall, this makes sense in all cases, as, for example, when markups are higher in the PCM method, they are also high in the production function method, and vice versa.

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